

# Ritabrata Chakraborty

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## EDUCATION

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### University of Pennsylvania

*M.S.E. in Robotics, GPA: -/4*

Pennsylvania, USA

Aug '26 – May '28

### Birla Institute of Technology and Science, Pilani (BITS Pilani)

*B.E. in Mechanical Engineering — Minor in Data Science, CGPA: 8.57/10*

Rajasthan, India

Oct '22 – May '26

## PUBLICATIONS & PEER REVIEWS

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### Journal Publications

- R. Chakraborty and K. Kishore, “*Navigating Without Satellites: A Critical Survey of GNSS-Denied UAV Navigation.*” Submitted to **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026. [🔗](#)
- R. Chakraborty and K. Kishore, “*Multi-Critic Off-Policy Reinforcement Learning for Quadrotor Velocity Control in Unknown Indoor Environments.*” Manuscript in preparation for **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026.

### Conference Publications

- Y. Wang, H. He, Y. Cao, J. Liang, R. Chakraborty, and G. Sartoretti, “*CogniPlan: Uncertainty-Guided Path Planning with Conditional Generative Layout Prediction.*” Accepted at **Conference on Robot Learning (CoRL)**, 2025; published in **Proceedings of Machine Learning Research (PMLR)**. [🔗](#)
- R. Chakraborty, T. Mian, and P. Kundu, “*An Efficient Approach for Synthetic Data Generation and Fault Diagnosis for Rotating Machinery.*” Accepted at **Prognostics and System Health Management Conference (PHM)**, 2025; published in **IET Conference Proceedings**. [🔗](#)
- K. Kishore and R. Chakraborty, “*Path Planning of Low-Altitude UAV for Tree Canopy Tracking and Orchard Monitoring.*” Publication awaiting intellectual property clearance, 2025. [🔗](#)

### Poster Presentations

- R. Chakraborty, “*High-Velocity Impact Testing and Analysis of Layered Metal-Composite Shields Using Electromagnetic Railgun.*” Presented as poster at **Pulsed Power Science, Technology and Application (PPSTA)**, Visakhapatnam, India, Sep 2024.

### Peer Reviews

- “*GS-SDE: 3D Gaussian Splatting-based Adaptive Scene Modeling in Semi-Dynamic Environments.*” Peer review completed for **IEEE Transactions on Automation Science and Engineering (T-ASE)**, 2026.
- “*Limits of Wearable Gait Analysis for Age Classification in Young Adults: Anthropometry Over Age.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- “*Generative AI and Computer Vision Based Multi-Assessment System for Early Prediction and Diagnosis of Autism Spectrum Disorder (ASD).*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- “*AquaNode: Design and Implementation of a Low-Cost Automated Water Tank Monitoring and Valve Control System Using Arduino and Ultrasonic Sensing.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- “*A research paper on IoT-Based Smart Irrigation Systems.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- “*Hybrid AI-Based Autonomous Drone Swarm Navigation.*” Peer review completed for **IEEE Gujarat Section Conference (GSC)**, 2026.
- “*Autonomous Robot Navigation for Smart Indoor using A\* and Deep Q-Network Model.*” Peer review completed for **IEEE Gujarat Section Conference (GSC)**, 2026.
- “*Deep Learning Driven Robot Simulation for Autonomous Navigation, Crop Health and Pesticide Spraying.*” Peer review completed for **IEEE Gujarat Section Conference (GSC)**, 2026.

RESEARCH EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                    |
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| <b>Research Intern</b><br><i>Supervisor: <a href="#">Dr. Avinash Gautam</a>, Associate Professor</i> <ul style="list-style-type: none"><li>Literature study of vision-language grounded multi-robot coordination and VLMaps-based navigation.</li></ul>                                                                                                                                                                                                             | <b>BITS Pilani</b><br><i>May '25 – Aug '25</i>     |
| <b>Research Intern</b><br><i>Supervisor: <a href="#">Dr. Kaushal Kishore</a>, Principal Scientist</i> <ul style="list-style-type: none"><li>Monocular vision-based UAV navigation for orchard monitoring using YOLOv11. <a href="#">↗</a></li><li>Multi-critic decomposition for off-policy reinforcement learning (TD3, SAC, DDPG) for autonomous indoor drone navigation. <a href="#">↗</a></li></ul>                                                             | <b>CSIR-CEERI</b><br><i>Jan '24 – May '26</i>      |
| <b>Research Intern</b><br><i>Supervisor: <a href="#">Dr. Guillaume Sartoretti</a>, Assistant Professor</i> <ul style="list-style-type: none"><li>Uncertainty-guided path planning via conditional generative layout prediction (CogniPlan). <a href="#">↗</a></li><li>Vision-attention-driven autonomous navigation with semantic understanding, extending <a href="#">CogniPlan</a> and <a href="#">ViNT</a>.</li></ul>                                            | <b>MARMoT Lab, NUS</b><br><i>Mar '25 – Dec '25</i> |
| <b>Research Assistant</b><br><i>Supervisor: <a href="#">Dr. Mani Shankar Dasgupta</a>, Associate Professor</i> <ul style="list-style-type: none"><li>Physics-based launcher mechanical design and FEM analysis. <a href="#">↗</a></li><li>Computer vision pipeline for 3D trajectory reconstruction and shot classification using <a href="#">PnLCalib</a>.</li><li>Finite element analysis of running track surfaces using hyperelastic material models.</li></ul> | <b>BITS Pilani</b><br><i>Apr '24 – Oct '25</i>     |
| <b>Research Intern</b><br><i>Supervisor: <a href="#">Dr. Pradeep Kundu</a>, Assistant Professor</i> <ul style="list-style-type: none"><li>Synthetic data generation for bearing fault diagnosis under data scarcity. <a href="#">↗</a></li><li>Physics-informed generative modeling for bearing fault diagnosis under data scarcity, validated on <a href="#">LiteFormer</a>.</li></ul>                                                                             | <b>KU Leuven</b><br><i>Sep '24 – May '26</i>       |
| <b>Research Assistant</b><br><i>Supervisor: <a href="#">Dr. Rajneesh Choubisa</a></i> <ul style="list-style-type: none"><li>Primal world beliefs study surveying engineering undergraduates on core worldview dimensions. <a href="#">↗</a></li></ul>                                                                                                                                                                                                               | <b>BITS Pilani</b><br><i>Aug '24 – Dec '24</i>     |
| <b>Research Intern</b><br><i>Supervisor: <a href="#">Dr. Rishi Verma</a>, Head (ESDS), PPEMD</i> <ul style="list-style-type: none"><li>Image-based visual servoing for automated radar control and UAV tracking using YOLOv8 and DeepSORT.</li><li>Impact testing and analysis of metal-composite shields using electromagnetic railguns.</li></ul>                                                                                                                 | <b>BARC</b><br><i>Jun '24 – Jul '24</i>            |

INDUSTRY EXPERIENCE

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| <b>ML Intern</b><br><i>Supervisors: <a href="#">Mr. Siddarth Malreddy</a>, Tech Lead Manager &amp; <a href="#">Mr. Ishan Nigam</a>, Senior ML Engineer</i> <ul style="list-style-type: none"><li>Object annotation automation and anomaly detection in <a href="#">uLabel</a>.</li></ul> | <b>Uber</b><br><i>Jul '25 – Dec '25</i>                 |
| <b>Subject Matter Expert</b><br><i>Self-supervised</i> <ul style="list-style-type: none"><li>Technical problem-solving and academic content authoring using MATLAB, CAD, and Python.</li></ul>                                                                                           | <b>Knowledge Cultivator</b><br><i>Nov '23 – Feb '24</i> |

PROJECTS

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| <b>Vision-Guided Robotic Arm for Simulated Oral Cancer Screening</b>                             | <i>Aug '25 – Dec '25</i> |
| <b>Autonomous Drone Navigation   MathWorks Global Student Drone Challenge 2025</b>               | <i>Mar '25 – Apr '25</i> |
| <b>Air Quality Index Forecasting Using Time Series Models <a href="#">↗</a></b>                  | <i>Nov '24 – Nov '24</i> |
| <b>3D Lunar Surface Modeling from DEM Data   ISRO Immersion Challenge 2024 <a href="#">↗</a></b> | <i>Jul '24 – Jul '24</i> |
| <b>Cricket Team Optimization   AMEX Campus Challenge 2024 (Product Track) <a href="#">↗</a></b>  | <i>Jun '24 – Jul '24</i> |
| <b>ExoMy Rover Navigation and UR3 Arm Motion Planning   ERC 2023 Remote <a href="#">↗</a></b>    | <i>Apr '23 – Sep '23</i> |

TEACHING EXPERIENCE

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| <b>Teaching Assistant</b><br><i>CIS 4190/5190: Applied Machine Learning, University of Pennsylvania</i> | <i>Aug '26 – Dec '26</i> |
| <i>CS F320: Foundations of Data Science, BITS Pilani</i>                                                | <i>Jan '26 – May '26</i> |
| <i>ME F218: Advanced Mechanics of Solids, BITS Pilani</i>                                               | <i>Jan '25 – May '25</i> |
| <i>ME F216: Materials Science and Engineering, BITS Pilani</i>                                          | <i>Sep '24 – Dec '24</i> |

HONOURS & SCHOLARSHIPS

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| <b>Institute Merit-cum-Need Scholarship — Consecutive Recipient (Top 8% of Students)</b> | <i>Oct '22 – May '26</i> |
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AWARDS & ACHIEVEMENTS

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| Finalist — Meta PyTorch OpenEnv Hackathon (India) <a href="#">🔗</a>                            | Apr '26 |
| 3 <sup>rd</sup> Place — MathWorks Global Drone Student Challenge 2025                          | Mar '25 |
| Finalist — ISRO Immersion Challenge (AI for Space & Geospatial Innovation) <a href="#">🔗</a>   | Jul '24 |
| Top 15 Overall, Top 5 in College — American Express Campus Challenge 2024 <a href="#">🔗</a>    | Jul '24 |
| 5 <sup>th</sup> Place & Best Maintenance Award — European Rover Challenge 2023 (Remote Finals) | Sep '23 |

LEADERSHIP EXPERIENCE

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| <b>Deputy Director of Academic Programming</b><br><i>Graduate and Professional Student Assembly (GAPSA), University of Pennsylvania</i>               | Aug '26 – Present |
| <b>Vice-President</b><br><i>South Asian Association (Rangoli), University of Pennsylvania</i>                                                         | Aug '26 – Present |
| <b>President &amp; Secretary</b><br><i>Mechanical Engineering Association (MEA), BITS Pilani</i>                                                      | Jun '24 – May '26 |
| <b>President &amp; Tech Fest Coordinator</b><br><i>Indian Society of Heating, Refrigerating, and Air Conditioning Engineers (ISHRAE), BITS Pilani</i> | Oct '24 – Jul '25 |
| <b>Project Manager</b><br><i>Tinkerer's Lab (TL), BITS Pilani</i>                                                                                     | May '24 – Jul '25 |
| <b>Treasurer</b><br><i>Bengali Cultural Association (Moruchhaya), BITS Pilani</i>                                                                     | May '24 – Jul '25 |

VOLUNTEERING EXPERIENCE

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| <b>Volunteer</b><br><i>Junoon, National Service Scheme (NSS), BITS Pilani</i> | Sep '23 – Sep '23 |
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PROFESSIONAL ACTIVITIES

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| <b>Graduate Student Member — IEEE</b> | 2026 |
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COMPETITIVE EXAMINATIONS

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| <b>GRE: 317 (Quantitative: 168, Verbal: 149, AWA: 3)</b>                      | 2025 |
| <b>TOEFL iBT: 108 (Reading: 27, Listening: 30, Speaking: 23, Writing: 28)</b> | 2025 |
| <b>JEE Advanced: All India Rank 14372</b>                                     | 2022 |

TECHNICAL SKILLS & COURSEWORK

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| <b>Relevant Coursework:</b> Machine Learning, Deep Learning, Foundations of Data Science, Applied Statistical Methods, Linear Algebra, Computer Programming, Engineering Optimization, Differential Equations, Design of Machine Elements, Digital Twins |
| <b>Programming Languages:</b> Python, C++, C, C#, Java, Bash (Shell)                                                                                                                                                                                     |
| <b>Robotics &amp; Simulation:</b> ROS (with Gazebo, Rviz), MAVROS, Navigation Stack, MoveIt, AirSim, Unity, Unreal Engine, MATLAB, Simulink, QGIS                                                                                                        |
| <b>AI &amp; Data Science:</b> PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, OpenCV, Open3D, NumPy, Pandas, SciPy, StatsModels, Matplotlib, Seaborn, Weights & Biases (W&B), Excel                                                         |
| <b>Cloud &amp; MLOps:</b> AWS (S3, EC2, SageMaker), Azure (Foundry), Google Cloud Platform (GCP), Docker, Git, GitHub Actions                                                                                                                            |
| <b>Hardware &amp; Embedded Systems:</b> NVIDIA Jetson (Nano, Orin), Raspberry Pi, Arduino, IMUs, Stereo Camera, 3D LiDAR                                                                                                                                 |
| <b>CAD &amp; Mechanical Simulation:</b> ANSYS Mechanical, SolidWorks, Fusion 360                                                                                                                                                                         |
| <b>Leadership &amp; Communication:</b> Event Planning, Team Management, Budget Allocation, Mentorship, Cross-Functional Coordination, Public Speaking, Stakeholder Outreach, Technical Writing                                                           |
| <b>Productivity Software:</b> Microsoft Office (Excel, PowerPoint, Word), Google Workspace                                                                                                                                                               |